

Practice Problems on the EM Algorithm

Advanced Statistical Methods II

Unless stated otherwise, write the observed data as $Y = y$, the missing or latent data as Z , the complete data as $X = (Y, Z)$, and

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y = y].$$

An EM cycle consists of an E-step followed by an M-step.

Problem 1: A genetics-type multinomial augmentation

Suppose

$$(X_1, X_2, X_3, X_4) \sim \text{Multinomial}\left(n; \frac{1}{2} + \frac{\theta}{4}, \frac{1-\theta}{4}, \frac{1-\theta}{4}, \frac{\theta}{4}\right), \quad 0 < \theta < 1.$$

For the observed counts, take

$$(x_1, x_2, x_3, x_4) = (125, 18, 20, 34), \quad n = 197.$$

- (a) Write the observed-data likelihood, up to a multiplicative constant not depending on θ .
- (b) Augment the first cell by writing $X_1 = Z_1 + Z_2$, where the augmented cell probabilities are $1/2$ and $\theta/4$. Write the complete-data log-likelihood, up to an additive constant.
- (c) Show that

$$Z_2 \mid X_1 = x_1, \theta^{(t)} \sim \text{Binomial}\left(x_1, \frac{\theta^{(t)}}{2 + \theta^{(t)}}\right).$$

Derive the E-step and the M-step.

- (d) Starting from $\theta^{(0)} = 0.5$, compute the first three EM updates. Also compute the observed log-likelihood (up to the same additive constant) at $\theta^{(0)}, \dots, \theta^{(3)}$ and verify that it increases.
- (e) Derive the likelihood equation for the direct MLE.

Problem 2: Missing observations from a normal sample

Let $Y_1, \dots, Y_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$, where σ^2 is known. Only $Y_1, \dots, Y_m$ are observed, with $m < n$; the remaining $Y_{m+1}, \dots, Y_n$ are missing.

- (a) Identify the missing data and write the complete-data log-likelihood for μ .
 (b) Derive $Q(\mu \mid \mu^{(t)})$ and show that the EM update is

$$\mu^{(t+1)} = \frac{\sum_{i=1}^m y_i + (n - m)\mu^{(t)}}{n}.$$

- (c) Show explicitly that

$$\mu^{(t+1)} - \bar{y}_{\text{obs}} = \left(1 - \frac{m}{n}\right) (\mu^{(t)} - \bar{y}_{\text{obs}}), \quad \bar{y}_{\text{obs}} = \frac{1}{m} \sum_{i=1}^m y_i,$$

and hence find the EM limit.

- (d) What does the contraction factor tell you about the speed of EM when the fraction of missing observations is large?

Problem 3: Gaussian mixture with known component parameters

Consider the two-component mixture

$$f(y \mid \pi) = \pi \phi(y; -1, 1) + (1 - \pi) \phi(y; 2.5, 1), \quad 0 < \pi < 1,$$

where $\phi(y; \mu, \sigma^2)$ denotes the $\mathcal{N}(\mu, \sigma^2)$ density. The observations are

$$-2.2, -1.1, 0.2, 2.4, 3.0.$$

Let $Z_i = 1$ if observation i came from component 1 and $Z_i = 0$ otherwise.

- (a) Write the complete-data log-likelihood for π .
 (b) Derive the responsibility

$$\gamma_i^{(t)} = \mathbb{P}(Z_i = 1 \mid y_i, \pi^{(t)})$$

and the M-step for π .

- (c) Starting from $\pi^{(0)} = 0.5$, calculate the five responsibilities at the first E-step and obtain $\pi^{(1)}$.
 (d) Perform two more EM updates. Comment on the ambiguous observation $y = 0.2$ versus the observations far inside either component.

Problem 4: Deriving the full two-normal-mixture EM updates

Let

$$f(y \mid \theta) = \pi \phi(y; \mu_1, \sigma_1^2) + (1 - \pi) \phi(y; \mu_2, \sigma_2^2),$$

where $\theta = (\pi, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2)$ and all five parameters are unknown. Introduce $Z_i \in \{0, 1\}$ as in Problem 3.

- (a) Write $\ell_c(\theta; y, z)$.
- (b) Derive the E-step responsibilities $\gamma_i^{(t)}$.
- (c) Starting from the Q -function, derive the M-step updates for π , μ_1 , μ_2 , σ_1^2 , and σ_2^2 .
- (d) Be explicit about which mean must appear in each variance update. Why is the denominator for the first variance $\sum_i \gamma_i^{(t)}$ rather than $n - 1$ or n ?
- (e) Give two practical pathologies of unconstrained Gaussian-mixture likelihood maximization and explain how they can affect EM.

Problem 5: Right-censored exponential lifetimes

Suppose $T_1, \dots, T_n$ are i.i.d. exponential with mean θ , so

$$f(t \mid \theta) = \frac{1}{\theta} e^{-t/\theta}, \quad t > 0.$$

The exact failure times of n_1 units are observed; for the remaining $n - n_1$ units, only right-censoring times c_j are observed, meaning $T_j \geq c_j$.

- (a) Treat the unobserved failure times of censored units as missing data. Write the complete-data log-likelihood.
- (b) Use the memoryless property to derive

$$\mathbb{E}(T_j \mid T_j \geq c_j, \theta^{(t)}) = c_j + \theta^{(t)}.$$

Hence derive the EM update.

- (c) Derive the observed-data likelihood directly and its MLE. Show that the fixed point of the EM update equals this MLE.
- (d) For exact failures 1.2, 2.0, 3.1 and two units censored at 2.5, start EM from $\theta^{(0)} = 2$. Compute five updates and compare them with the direct MLE.

Problem 6: Two life-testing experiments with only status indicators

Lifetimes are exponential with unknown mean θ . In experiment I, exact lifetimes $y_1, \dots, y_N$ are observed. In experiment II, M new units are inspected only once, at time t . For each unit in experiment II we record

$$E_i = \begin{cases} 1, & \text{unit is still functioning at time } t, \\ 0, & \text{unit has failed by time } t. \end{cases}$$

Let $Z = \sum_{i=1}^M E_i$.

- (a) If X_i denotes the unobserved lifetime in experiment II, derive

$$\mathbb{E}(X_i \mid E_i = 1, \theta) = t + \theta$$

and

$$\mathbb{E}(X_i \mid E_i = 0, \theta) = \theta - \frac{te^{-t/\theta}}{1 - e^{-t/\theta}}.$$

- (b) Write the complete-data log-likelihood for the combined experiments and derive the EM update map $\theta^{(t+1)} = F(\theta^{(t)})$.
- (c) Take $y = (1.2, 2.0, 2.8, 3.5)$, $M = 6$, inspection time $t = 2.5$, and suppose $Z = 2$ units are still functioning. Starting from $\theta^{(0)} = 2$, compute at least four EM updates.
- (d) Explain why the E-step here is richer than simply replacing each censored lifetime by t .

Problem 7: Parameter-dependent support (tricky)

Now replace the exponential lifetime model in Problem 6 by

$$X \sim \text{Uniform}(0, \theta], \quad \theta > 0.$$

Suppose the same two-experiment design is used, and at least one unit in experiment II survives beyond inspection time t .

- (a) A tempting “EM-like” rule is to replace a missing X_i by its conditional mean and then plug these means into the complete-data MLE $\max\{Y_{\max}, X_{\max}\}$. Show that this produces the iteration

$$\theta^{(t+1)} = \max \left\{ Y_{\max}, \frac{t + \theta^{(t)}}{2} \right\}.$$

What fixed point does this suggest?

- (b) Derive the observed-data likelihood when $Z \geq 1$ units survive past t , and show that, away from the boundary imposed by $Y_{\max}$, its interior maximizer is

$$\bar{\theta} = \frac{N + M}{N + M - Z} t.$$

- (c) Explain precisely why the plug-in iteration in part (a) is not a valid EM algorithm.
- (d) Suppose EM is initialized at $\theta^{(0)} > \max\{Y_{\max}, t\}$. Show that the genuine Q -function satisfies

$$Q(\theta \mid \theta^{(0)}) = \begin{cases} -(N + M) \log \theta + C, & \theta \geq \theta^{(0)}, \\ -\infty, & 0 < \theta < \theta^{(0)}, \end{cases}$$

for a constant C independent of θ . What does one EM update do?

Problem 8: The ascent property of EM

Let

$$p_t(z) = p(z \mid y, \theta^{(t)}), \quad Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{p_t}[\log p(y, Z \mid \theta)].$$

Assume all quantities below are well defined.

(a) Show the identity

$$\ell(\theta) - \ell(\theta^{(t)}) = Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}) + D_{\text{KL}}(p_t \parallel p(\cdot \mid y, \theta)),$$

where $\ell(\theta) = \log p(y \mid \theta)$.

(b) Deduce the ascent property

$$Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)}) \implies \ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).$$

(c) Does monotone increase of the likelihood imply convergence to the global maximum? Explain briefly.

Problem 9: A Bernoulli mixture model

For $n = 1, \dots, N$, observe a binary vector

$$Y_n = (Y_{n1}, \dots, Y_{nM}) \in \{0, 1\}^M.$$

Assume a K -component mixture. Conditional on latent class $Z_n = k$, the M coordinates are independent Bernoulli variables with

$$\mathbb{P}(Y_{nm} = 1 \mid Z_n = k) = p_{km},$$

and $\mathbb{P}(Z_n = k) = \pi_k$, where $\sum_{k=1}^K \pi_k = 1$.

- (a) Write the component probability $f_k(y_n)$ and the complete-data log-likelihood using indicators $Z_{nk} = \mathbf{1}\{Z_n = k\}$.
- (b) Derive the posterior class responsibilities $\gamma_{nk}^{(t)}$.
- (c) Derive the M-step updates for π_k and p_{km} .
- (d) Explain in one or two sentences why the Bernoulli-mixture M-step has the same “expected counts divided by expected totals” structure as the Gaussian-mixture M-step.

References

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